

DTSA 5703

Parallel Computing with MPI

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (84.6 / 100)	71.3 / 100
SPECIALIZATION Introduction to High Performance and Parallel Computing		

71

CURRENT AVG SCORE
across 6 job roles

95

POTENTIAL SCORE
with recommended additions

+32

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Provides a high-level introduction to modern computer node architectures of high-performance and cloud computing instances. This course is targeted to scientists, engineers, scholars, or anyone seeking to develop the software that needs to run efficiently on high-performance computing environments or in the cloud. For this course, students need basic programming skills in either C++, Fortran 90, or Python in addition to basic Linux skills. Students completing this course will have a basic understanding of how to find bottlenecks in their programs as well as how to address those bottlenecks. The course will provide a high-level introduction to modern compute node architectures of high-performance and cloud computing instances.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	84.6	100.0	+15.4	MEDIUM
AI Engineer	82.3	100.0	+17.7	MEDIUM
MLOps Engineer	76.4	100.0	+23.6	MEDIUM
Applied Scientist	68.3	100.0	+31.7	HIGH
ML Engineer	65.5	90.0	+24.6	MEDIUM
Data Scientist	50.5	82.5	+32.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Model Deployment & Serving 2 of 6 roles

production

Deep Learning 1 of 6 roles

cnn

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** Add **System Design** coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- MEDIUM** Consider **Model Deployment & Serving** content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

- LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: cnn.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

84.6 → 100.0 (+15.4 pts)

data engineering learning tools solutions design science models

AI Engineer

82.3 → 100.0 (+17.7 pts)

data robotics engineering models learning infrastructure technical code

MLOps Engineer

76.4 → 100.0 (+23.6 pts)

infrastructure detection data production systems learning machine media

Applied Scientist

68.3 → 100.0 (+31.7 pts)

computational literature problems models technical data learning systems

ML Engineer

65.5 → 90.0 (+24.6 pts)

cnn media data learning machine systems models production

Data Scientist

50.5 → 82.5 (+32.0 pts)

data characteristics scripting different social engineering people problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.